A monolithically printed cell for topology-programmable soft robotic surfaces and bodies

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Soft robots can achieve compliant motion, but most are fabricated as bodies whose size, topology, and function are fixed once made (*1–3*). A general route to adaptable soft robots is to build from repeated actuated units that can be assembled into different topologies while preserving continuous deformation (*4–7*). Here we show a topology-programmable soft robotic architecture based on a single monolithically printed, pneumatically actuated cellular unit that integrates actuation, compliant linkage motion, and modular connectivity. In planar arrays, pressure-driven cell expansion and module-scale bending bias produce programmable three-dimensional surface morphing, including multi-feature shape display, overhang generation, and object interaction. Reconfiguring the same building block into cylindrical topologies yields continuum-style bending and grasping, peristaltic locomotion, and self-rolling locomotion. The results show how changing topology and actuation pattern, rather than redesigning the cellular unit for each task, allows one manufacturable architecture to generate multiple soft robotic morphologies and functions.

Introduction

Soft robots are often fabricated as task-specific bodies whose size, connectivity, and morphology are fixed by design. When the required task or body plan changes, adaptation usually requires redesign and refabrication. A more scalable route is a soft robotic architecture built from repeated units that can be manufactured once and assembled into different topologies; in this route, the same local

actuator is reused while module connectivity and actuation pattern set the global surface or body response (*1–3*).

This goal remains difficult because the architecture must combine continuous shape change, distributed actuation, scalable fabrication, and topology-level reassembly (*1*, *4*). Voxel-based soft robots and modular soft building blocks show how compliant robots can be assembled and reassembled from repeated components (*6–8*). Compact soft surfaces and shape displays show how dense arrays can render height fields, tune stiffness, and interact with objects (*9–17*). Soft continuum robots can bend, grasp, and locomote (*18–24*). The remaining gap is a single soft robotic building block that is locally actuated, manufactured as a monolithic pneumatic unit, assembled as a continuous programmable surface, and then reorganised into non-planar robot bodies without redesigning the actuator.

A route to this system is to encode shape-generating behaviour in the geometry of a tessellated cell. Architected materials and buckling-based systems show that geometry can program deformation and stiffness, but they often lack the local controllability needed for multifunctional robotic behaviour (*25–28*). We therefore sought a cellular unit that is manufacturable, pneumatically actuated, tessellatable, and able to convert local expansion into controlled out-of-plane morphology. The Sarrus linkage provides a useful basis because lateral expansion is accompanied by leg rotation and geometry-dependent mechanical response (*29–33*).

Here we introduce a topology-programmable soft robotic architecture based on one monolithically 3D-printed, pneumatically actuated cellular unit with integrated actuation and modular connectivity. The key advance is the use of one repeated pneumatic cell as both a programmable surface primitive and a reassemblable robot-body primitive, linking capabilities that are usually developed in separate systems. Planar assemblies generate continuous surface morphing, multi-feature shape display, overhang formation, and object interaction. Cylindrical assemblies generate continuum-style bending, grasping, peristaltic locomotion, and self-rolling locomotion. The same manufacturable cellular architecture therefore supports multiple larger-scale robotic functions by changing topology and actuation pattern, rather than redesigning the cellular actuator and connection scheme for each task.

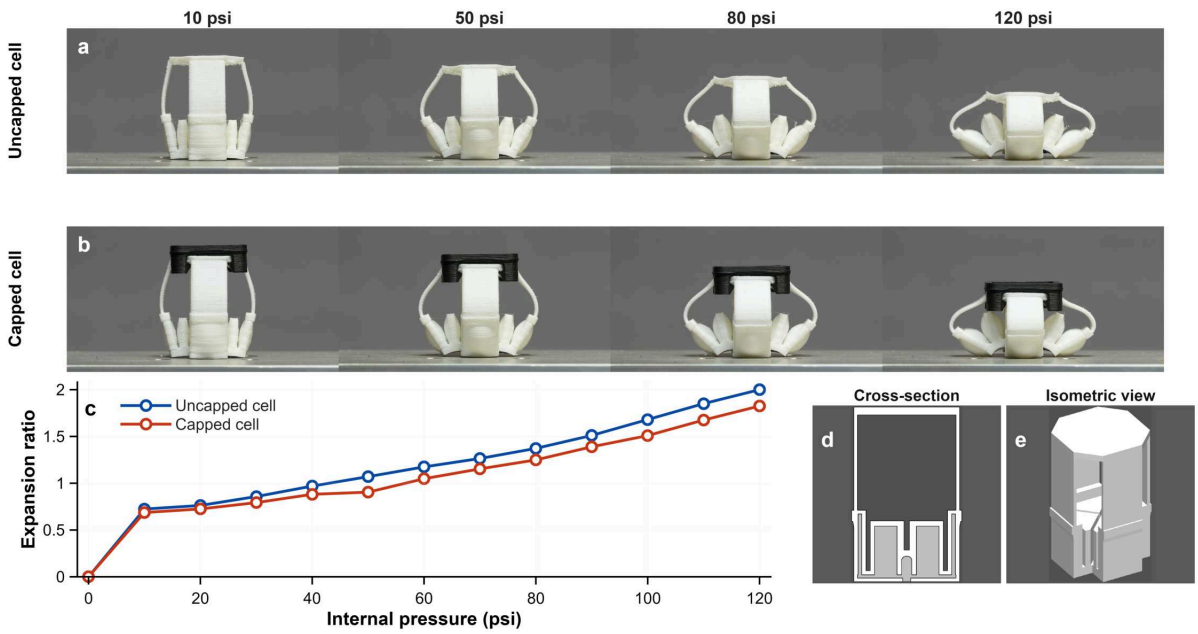

Figure 1. Cell geometry and pressure-driven expansion. (a,b) Uncapped and capped cells across the actuation range. (c) Pressure-dependent lateral expansion ratio for both cell variants. (d) Cell cross-section showing the compliant Sarrus linkage and integrated PneuNet actuator. (e) Isometric view of the cell geometry.

Results

The Results are organised around three explicit levels of the architecture: the cell/module level, where pressure-driven expansion and bending bias are established; the planar-array level, where actuation patterns generate programmable surface features and object interactions; and the cylindrical-robot level, where the same modules are reassembled into bending, grasping, peristaltic, and rolling bodies.

A monolithic cell converts pressure into expansion and biased bending

The unit cell integrates a compliant linkage geometry and pneumatic actuator in a single printed part (Fig. 1). Pressurization expands the cell laterally, while the cap reduces open gaps at the top surface during object interaction. The cell therefore provides a repeated expansion primitive that can be assembled into larger soft robotic architectures.

Grouping four cells into a 2×2 module reduces tubing complexity and introduces a directional bending bias (Fig. 2). The plotted bending angles, θ_s for a single module and θ_d for a two-module

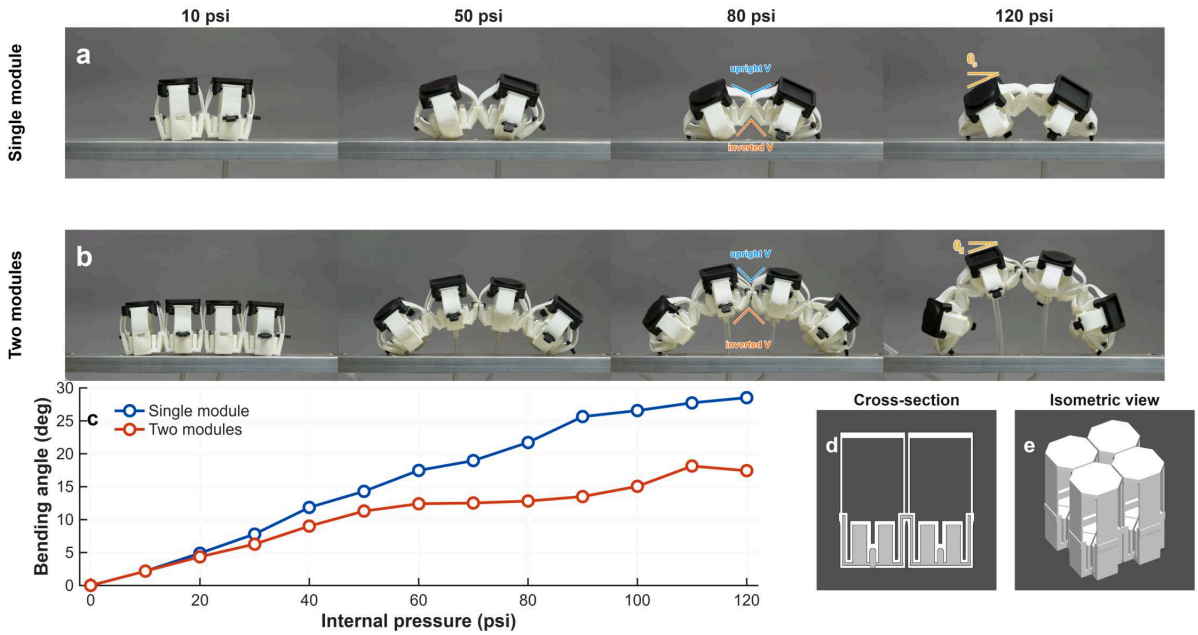

Figure 2. Module-level directional bending. (a,b) Single-module and two-module configurations across the actuation range. Upright-V and inverted-V linkage leg pairs are indicated on the 80 psi images. (c) Measured bending angles, θ_s and θ_d, for the single- and two-module configurations. (d) Module cross-section showing the cross-cell pneumatic channels. (e) Isometric view of the module geometry.

assembly, quantify the module tilt produced by pressure-driven deformation. In the actuated images, the upright-V and inverted-V linkage leg pairs indicate how local leg motion biases the module toward a repeatable convex bending direction. This gives each module both an expansion mode and a preferred out-of-plane response.

Mechanical tests show that pressure-driven deformation is accompanied by pressure-dependent load response and hysteresis (Fig. 3). Uncapped cells, capped cells, and 3×3 arrays all show increasing force capacity with internal pressure. The array response is not a simple scaled copy of the single-cell response: force, stiffness, and hysteresis depend on the coupled deformation of neighbouring cells and module connections. These measurements define the mechanical context for the surface and robot demonstrations that follow.

Planar topology programs continuous surface features

In planar arrays, actuation patterns convert local module expansion into continuous three-dimensional surface features (Fig. 4). Increasing the actuated footprint increases dome height until the response

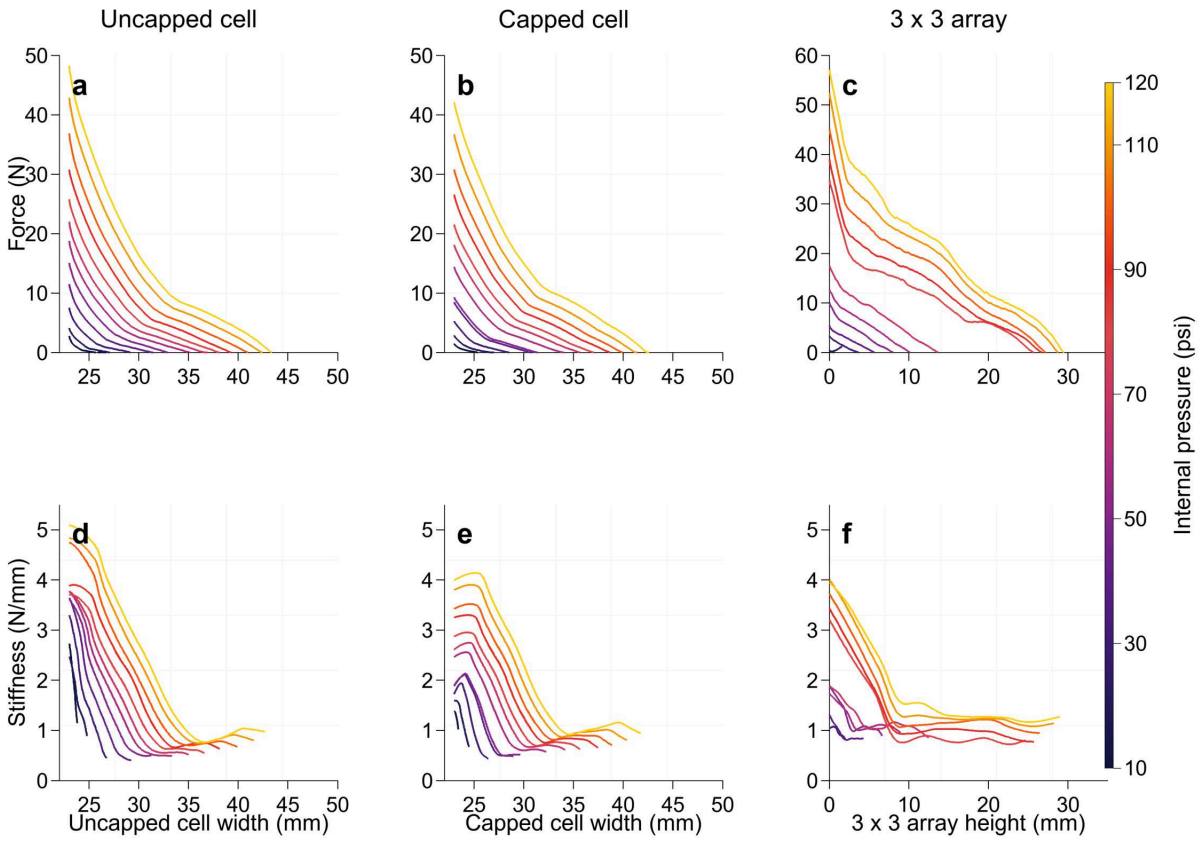

Figure 3. Pressure-dependent force and stiffness. (a–c) Measured force response of the uncapped cell, capped cell, and 3×3 module array over the compression stroke. (d–f) Local stiffness computed from the corresponding force-displacement curves. Colour indicates internal pressure.

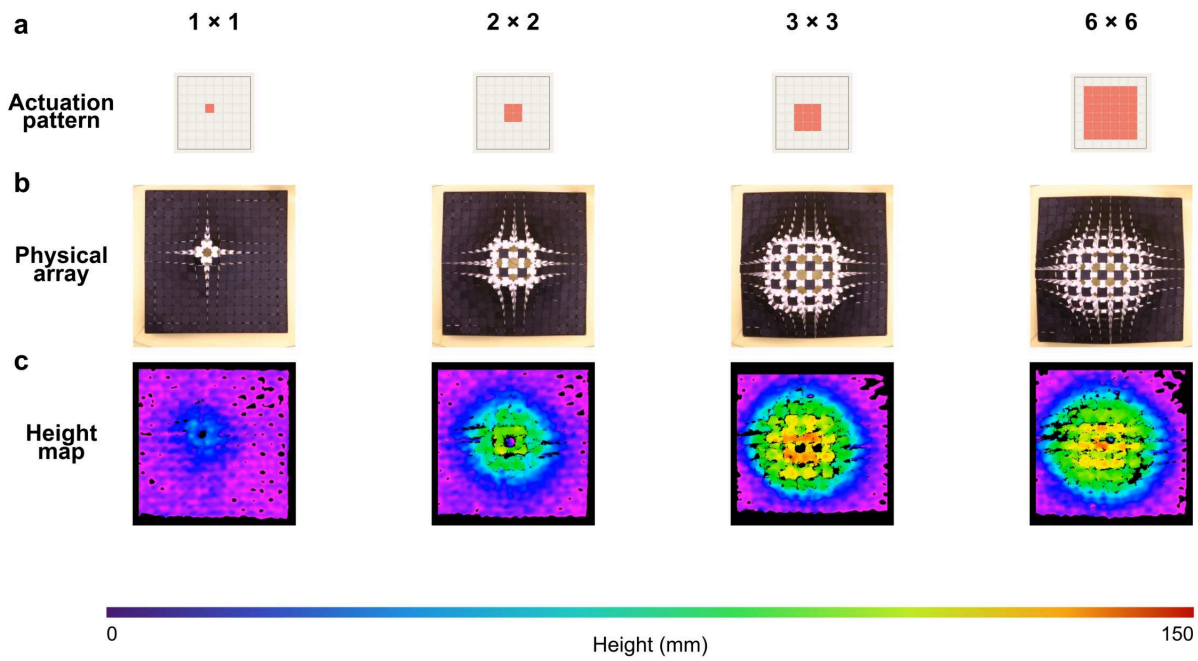

Figure 4. Planar single-dome feature scaling. (a) Actuation patterns for increasing group size. (b) Physical array images showing the corresponding surface features. (c) Height maps of the same states. Increasing the actuated footprint increases feature height until the response begins to saturate; double-dome, overhang, object-interaction, and travelling-wave planar demonstrations are provided in the supplementary information.

begins to saturate. Additional planar demonstrations in the supplementary information show double-dome spatial resolution, bilayer overhang generation, object interaction, and travelling waves. Together, these results show that the architecture behaves as a mechanically coupled surface: actuation is local, but the resulting shape is set by the interaction between neighbouring modules, boundary conditions, and topology.

Cylindrical topology converts the same module into robot bodies

Rolling a module sheet into a tube changes the available deformation modes without changing the cellular unit (Fig. 5). The cylindrical topology can be used as a simple continuum arm that opens, closes, transports, and releases an object. Related C-curve and S-curve configurations are provided in the supplementary information. Thus, topology-level reconfiguration changes the function from a shape display surface to a soft robot body.

Sequential actuation of cylindrical bodies produces locomotion (Fig. 6). A crawler uses a propagating

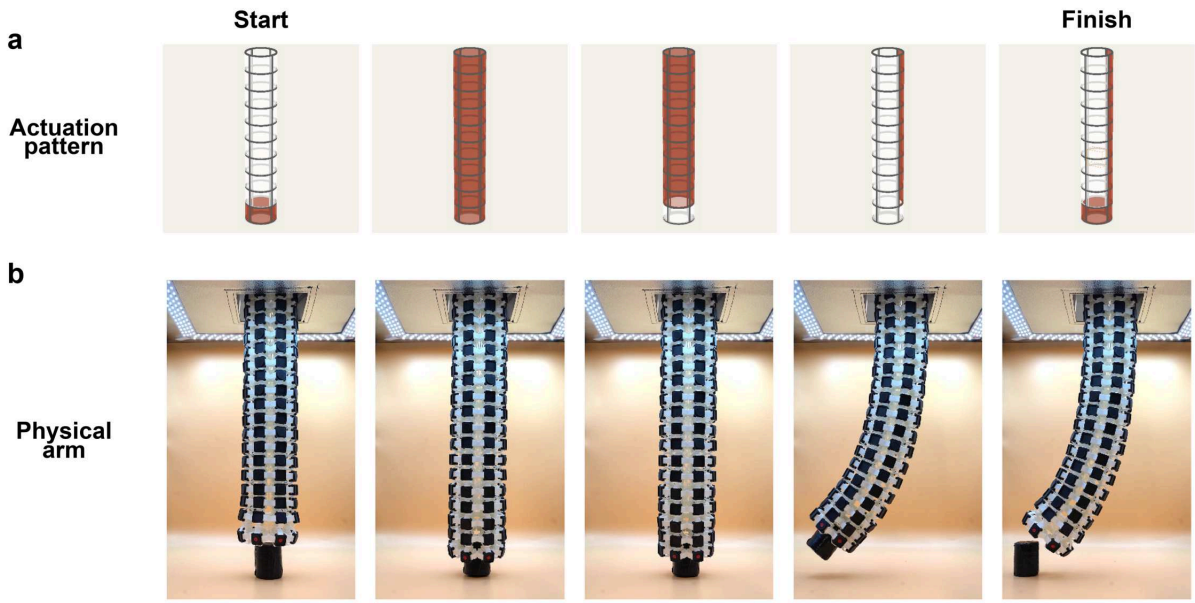

Figure 5. Continuum-arm object manipulation using the cylindrical topology. (a) Actuation-pattern sequence. (b) Physical arm images showing the corresponding manipulation states. The distal ring opens to receive the object, the body lowers around it, the ring closes to grasp it, and a C-curve actuation pattern transports and releases the object at a new location.

actuation sequence to generate repeated forward motion over multiple gait cycles. A wheel-like self-rolling assembly is provided in the supplementary information. These demonstrations show that the same cell architecture can move beyond static reconfiguration to robot-scale motion.

Discussion

This work establishes a topology-programmable soft robotic architecture in which one monolithically printed pneumatic unit can be reassembled into distinct robotic topologies. The central result is that local pressure-driven expansion can be converted by module-scale bias into controlled three-dimensional deformation. That mechanism allows the same repeated unit to support planar surface morphing, object interaction, continuum-style bending, grasping, peristaltic locomotion, and self-rolling locomotion. Relative to modular soft building-block robots and voxel robots, the system emphasises continuous, locally addressable shape formation from a monolithic pneumatic cell; relative to soft reconfigurable surfaces, it also functions as a reassemblable robot body (*6–9*). In this sense, the function of the system is reassigned by changing assembly topology and actuation

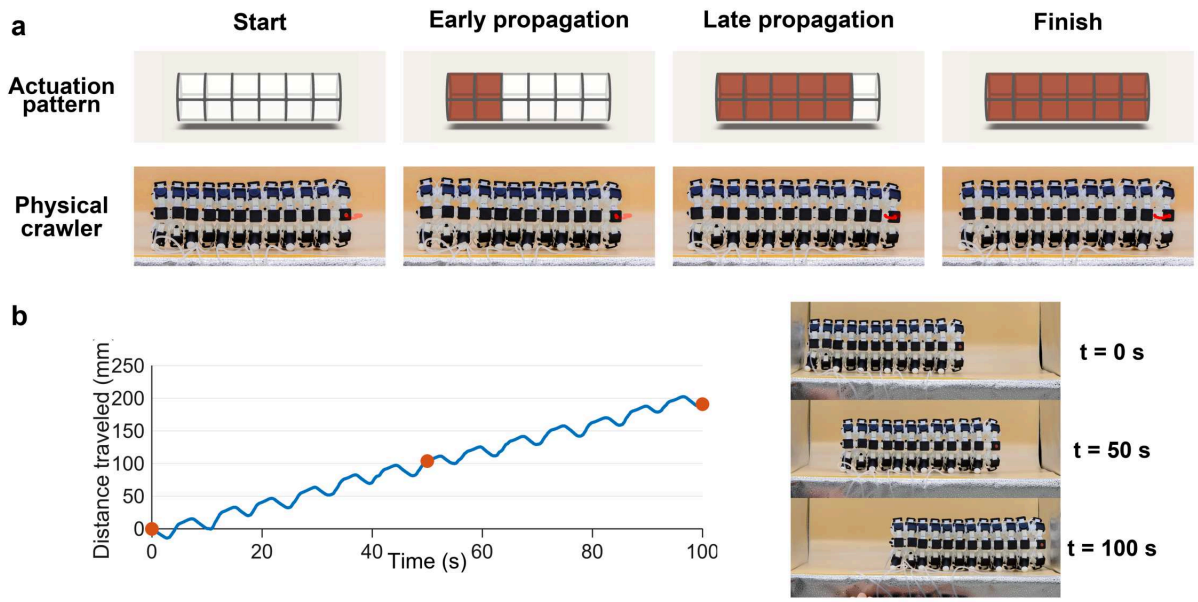

Figure 6. Peristaltic locomotion from the cylindrical topology. (a) Actuation-pattern sequence and corresponding physical crawler states. (b) Forward displacement over time with representative snapshots. Sequentially actuating six groups creates a travelling deformation wave that advances the crawler over multiple gait cycles; self-rolling locomotion is provided in the supplementary information.

pattern.

The experiments also define practical limits. In planar arrays, spatial resolution and amplitude are coupled because neighbouring modules share deformation through the continuous sheet. Larger spacings of unactuated cells improve feature separation but reduce peak height. In cylindrical bodies, topology unlocks new functions but currently requires tethered pneumatic inputs and manual assembly. The next step is integrated robotic control: electronically addressed actuation, sensing of external force and curvature, and vision-based feedback could allow the body state to be inferred and regulated in real time. More broadly, integrated sensing, closed-loop control, and algorithmic pattern planning would allow the same cellular architecture to move from proof-of-principle demonstrations toward autonomous soft robotic systems that can be assembled, programmed, sensed, controlled, and repurposed for changing tasks.

Materials and methods summary

Cells were fabricated as monolithic soft pneumatic units that combine a compliant Sarrus-linkage-inspired geometry with integrated PneuNet actuation. Four cells were connected into 2×2 modules with shared pneumatic channels and snap-fit connection features. Mechanical characterization measured pressure-dependent expansion, force-displacement response, hysteresis, and local stiffness. Planar arrays were assembled from repeated modules and actuated with prescribed pressure patterns. Cylindrical bodies were formed by rolling module sheets into tubes or wheel-like assemblies and driving selected module groups sequentially. Detailed fabrication, measurement, and supplementary figure information are provided in the supplementary information.

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Author contributions

A.N.P. performed the research, analyzed the data, prepared the figures, and drafted the manuscript. P.M.R. and C.D.O. supervised the work and revised the manuscript.

Competing interests

The authors declare no competing interests.

Data and materials availability

All data needed to evaluate the conclusions are included in the paper and supplementary information. Raw data, analysis scripts, and figure-generation code should be deposited in a public repository before final submission or made available from the corresponding author if repository release is not possible before review.

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